

# Towards a Living Benchmark for Historical HTR



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## One error rate hides three different mistakes.



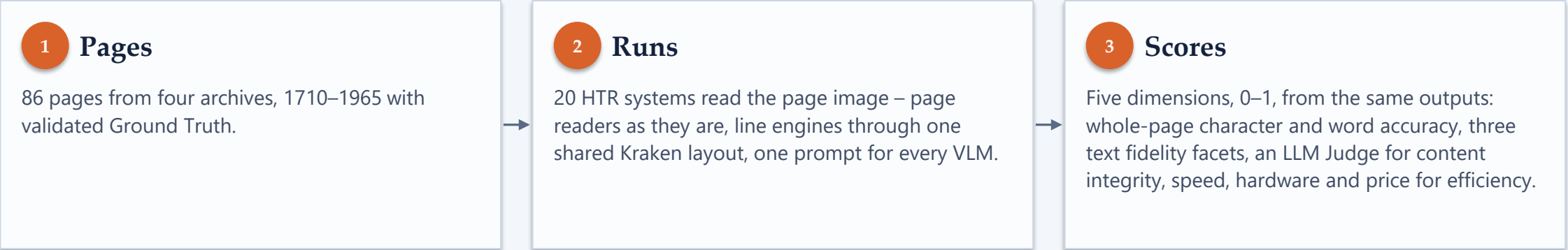
Ground Truth *die in Deutschland nur bei Harzburg am Harze entdeckt*  
Gemini 3.5 Flash *die in Deutschland nur bei Harzburg am **Wege** entdeckt*  
Transkribus Text Titan II *die in Deutschland **im** bei **Herzberg** am **Hage** entdeckt.*

- Wege** a different word in place of "Harze"
- Herzberg** a real, but wrong town 60 km away
- entdeckt** modernised spelling

These errors differ in gravity. Orthographic variation does not influence understanding and meaning.  
**A plausible, yet wrong entity transcription is more drastic.**

## We need a benchmark for historical document HTR that goes beyond CER.

### How this benchmark works



## The material: 4 corpora of German handwriting, 1710–1965

**Ratsprotokolle 1710–1803**  
Kurrent + Antiqua: council minutes; diplomatic transcription  
34 pages; 870 lines

**Humboldt journals 1790–1805**  
Kurrent / Latin: travel journal; scholarly edition EHD  
20 pages; 602 lines

**Forest operates 1857–1895**  
Kurrent: forestry reports and tables; corrected machine output  
22 pages; 823 lines

**Laubmann journals 1917–1965**  
Sütterlin era; field journals; corrected machine output  
10 pages; 338 lines

### D1 Character accuracy

How much of the page came back right?

**Transkribus German Genius gets 82.0 % of the page's characters right, PaddleOCR-VL 1.6 33.8 %.**

reference *Concl. Sub Consuetis clausul-*  
Gemini 3.5 Flash **Concl.**: Sub Consuetis **clausulis** CER 14 %  
TrOCR Kurrent XVI-XVII **Concl.**: **sub consuetis clausulis** CER 29 %  
Transkribus Text Titan II **Concl.**: Sub Consuetis **clausus** CER 11 %

**Measured** 1 – CER over the whole page text.

### D2 Word accuracy

How many of the page's words came back right?

**Gemini 3.5 Flash gets 62.5 % of the page's words right, PaddleOCR-VL 1.6 6.5 %. One wrong character costs the whole word.**

reference *Concl. Sub Consuetis clausul-*  
Gemini 3.5 Flash **Concl.**: Sub Consuetis **clausulis** WER 50 %  
TrOCR Kurrent XVI-XVII **Concl.**: **sub consuetis clausulis** WER 100 %  
Transkribus Text Titan II **Concl.**: Sub Consuetis **clausus** WER 50 %

**Measured** 1 – WER over the whole page text.

### D3 Diplomatic fidelity

Is the text kept as written – letter forms, punctuation, capitals?

**No generalist keeps the letter forms: glyph F1 0.35 (Gemini 3.5 Flash) to 0.00 (olmOCR-2 7B).**  
Punctuation survives for 77–37 % of the marks, capitals for 96–85 %.

**Measured** mean of: period-glyph F1 (Ratsprotokolle); punctuation kept in place; capitalisation kept.

### D4 Content integrity

Do the mistakes change the meaning – would a reader be misled?

**Of the lines with errors, the judge rated 23 % misleading for Gemini 3.5 Flash, 56 % for olmOCR-2 7B.** Numbers: Gemini 3.5 Flash keeps 81 %, Kraken 19th-c. Kurrent 22 %.

**Measured** mean of: 1 – share of judged lines rated misleading (LLM judge); numeral F1 (dates, sums, entry numbers).

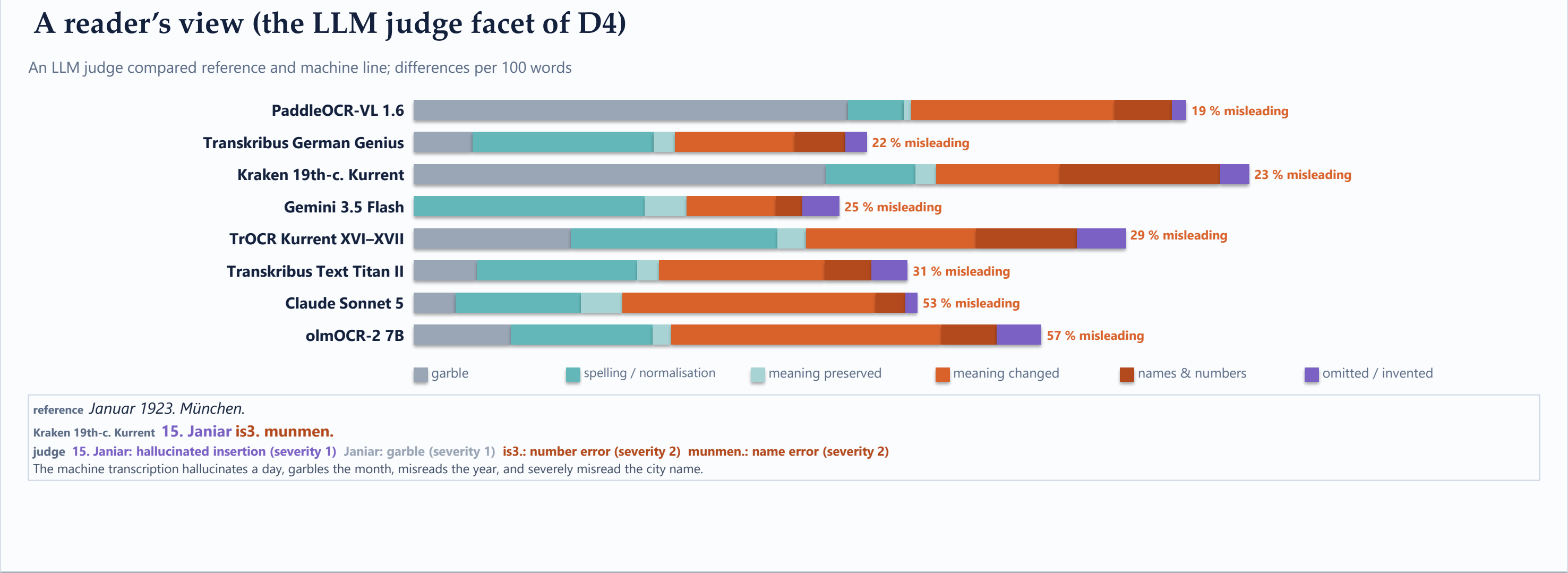
### D5 Efficiency

What does a page cost – in time, hardware and money?

|                           | speed  | hardware   | price  | D5   |
|---------------------------|--------|------------|--------|------|
| Kraken 19th-c. Kurrent    | 1.3 s  | laptop CPU | 0.01 c | 0.97 |
| TrOCR Kurrent XVI-XVII    | 4.2 s  | GPU 8 GB   | 0.01 c | 0.79 |
| PaddleOCR-VL 1.6          | 127 s  | GPU 8 GB   | 0.01 c | 0.67 |
| olmOCR-2 7B               | 13.3 s | GPU 24 GB  | 0.01 c | 0.59 |
| Gemini 3.5 Flash          | 3.9 s  | API        | 0.6 c  | 0.56 |
| Claude Sonnet 5           | 14.2 s | API        | 1.5 c  | 0.39 |
| Transkribus German Genius | 11.1 s | web app    | 26 c   | 0.31 |
| Transkribus Text Titan II | 14.8 s | web app    | 26 c   | 0.28 |

**Kraken 19th-c. Kurrent: 1.3 s a page on a laptop CPU, no fee. Transkribus Text Titan II: 14.8 s and 26 ¢ a page.** Only on speed does a hosted system come close: Gemini 3.5 Flash, 3.9 s.

**Measured** 1/3 speed + 1/4 hardware + 1/4 price · log scales 1–100 s and 0.1 ¢–1 \$ · hardware CPU 1; GPU 8 GB 0.8; 24 GB 0.5; hosted 0.1; open weights priced by electricity.



## Every system on every corpus by dimension

|                           | Ratsprotokolle |     |     |     |     | Humboldt |     |     |     |     | Forest operates |     |     |     |     | Laubmann |     |     |     |     |
|---------------------------|----------------|-----|-----|-----|-----|----------|-----|-----|-----|-----|-----------------|-----|-----|-----|-----|----------|-----|-----|-----|-----|
|                           | D1             | D2  | D3  | D4  | D5  | D1       | D2  | D3  | D4  | D5  | D1              | D2  | D3  | D4  | D5  | D1       | D2  | D3  | D4  | D5  |
| Gemini 3.5 Flash          | .80            | .52 | .64 | .70 | .57 | .68      | .47 | .88 | .80 | .57 | .89             | .77 | .92 | .84 | .56 | .94      | .84 | .92 | .86 | .44 |
| Transkribus German Genius | .88            | .63 | .52 | .82 | .31 | .66      | .38 | .86 | .71 | .31 | .85             | .68 | .82 | .69 | .31 | .88      | .64 | .83 | .82 | .31 |
| Transkribus Text Titan II | .83            | .54 | .47 | .71 | .28 | .61      | .29 | .81 | .66 | .28 | .82             | .61 | .75 | .62 | .28 | .86      | .62 | .86 | .81 | .28 |
| Claude Sonnet 5           | .66            | .33 | .54 | .60 | .39 | .51      | .21 | .76 | .60 | .41 | .67             | .43 | .80 | .70 | .39 | .89      | .72 | .89 | .79 | .36 |
| TrOCR Kurrent XVI-XVII    | .80            | .31 | .46 | .58 | .82 | .44      | .05 | .67 | .43 | .79 | .74             | .31 | .64 | .43 | .78 | .62      | .10 | .70 | .47 | .76 |
| Kraken 19th-c. Kurrent    | .55            | .10 | .49 | .14 | .98 | .39      | .04 | .68 | .22 | .98 | .66             | .23 | .65 | .24 | .95 | .57      | .13 | .76 | .26 | .89 |
| olmOCR-2 7B               | .54            | .20 | .45 | .53 | .61 | .20      | .06 | .60 | .35 | .64 | .27             | .09 | .58 | .50 | .58 | .61      | .36 | .73 | .78 | .57 |
| PaddleOCR-VL 1.6          | .38            | .08 | .42 | .22 | .68 | .20      | .04 | .63 | .34 | .52 | .29             | .04 | .58 | .33 | .67 | .54      | .12 | .75 | .50 | .65 |